

Q1.

Check for homogeneity of equations. An inhomogeneous equation is an incorrect equation.

Let base units be represented by BU.

A: LHS: $BU = \text{kg m}^2 \text{s}^{-2}$

RHS: $BU = (\text{kg m s}^{-1})(\text{m s}^{-1}) = \text{kg m}^2 \text{s}^{-2}$

LHS = RHS $\Rightarrow$ homogeneous equation (wrong answer)

B: LHS : $BU = (\text{kg m}^2 \text{s}^{-2})^2 = \text{kg}^2 \text{m}^4 \text{s}^{-4}$

RHS: BU of $p^2 c^2 = (\text{kg m s}^{-1})^2 (\text{m s}^{-1})^2 = \text{kg}^2 \text{m}^4 \text{s}^{-4}$

BU of $m^2 c^4 = \text{kg}^2 (\text{m s}^{-1})^4 = \text{kg}^2 \text{m}^4 \text{s}^{-4}$

All terms on both sides of the equal sign have the same units $\Rightarrow$ equation is homogeneous (wrong answer)

C: LHS: $BU = \text{kg m}^2 \text{s}^{-2}$

RHS: $BU = (\text{kg m s}^{-1})^2 (\text{m s}^{-1}) = \text{kg m}^3 \text{s}^{-3} \Rightarrow$ inhomogeneous equation (correct answer)

D: LHS: $BU = \text{kg m}^2 \text{s}^{-2}$

RHS: $BU = \frac{(\text{kg m s}^{-1})^2}{\text{kg}} = \text{kg m}^2 \text{s}^{-2} \Rightarrow$ homogeneous equation (wrong answer)

Ans: C

Q2